VERIQTA

SIEM FROM ZERO



CLEAR EXPLANATIONS
Beginner friendly, practical and accurate.

REAL EXAMPLES
Hands-on labs and real world scenarios.

PRACTICAL SKILLS
Build confidence in monitoring, detection and investigation.

BEGINNER TO PRACTICAL
Step by step from fundamentals to a complete investigation.

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What Is SIEM?

1. WHAT IS SIEM?

SIEM stands for Security Information and Event Management.
It is a centralized platform that collects, stores, analyzes and correlates security logs from different systems to detect threats and support incident response.

2. WHY DO ORGANIZATIONS USE SIEM?

- Gain visibility across all systems and networks
- Detect suspicious and malicious activity
- Identify security threats early
- Support investigation and incident response
- Meet security and compliance requirements
- Monitor users, applications and infrastructure in real time

3. HOW SIEM WORKS (HIGH LEVEL FLOW)

Collect Logs
From servers, endpoints, firewalls, cloud and applications

→

Process and Store
Normalize and index logs for analysis

→

Analyze and Correlate
Look for patterns, suspicious activity and threats

→

Generate Alerts
Create alerts for potential security issues

→

Investigate and Respond
Security teams investigate and take action

4. WHO USES SIEM?

- Security Operations Center (SOC) analysts
- IT and security teams
- System administrators
- Compliance and audit teams
- Organizations of all sizes
- Cloud and infrastructure teams

5. EXAMPLES OF LOG SOURCES

- Servers (Windows/Linux)
- Endpoints
- Firewalls
- Cloud Platforms (AWS, Azure, GCP)
- Applications (Web, Database)
- Identity Systems (Active Directory, Entra ID)
- Network Devices (Routers, Switches)
- Databases (MySQL, PostgreSQL)

6. KEY POINT TO REMEMBER

SIEM does not stop attacks automatically. It helps security teams detect, understand and respond to threats faster.

7. JUNIOR ENGINEER TIP

Think of SIEM as a single place where all security logs come together, so you can see what is happening across your entire environment.

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How SIEM Works

1. OVERVIEW

A SIEM collects logs from different systems, processes them, looks for suspicious activity, generates alerts and helps security teams investigate and respond to threats.

2. WHY IT MATTERS

- Brings all security data into one place
- Helps detect threats early
- Provides real-time monitoring
- Supports incident response
- Improves visibility and compliance

3. SIEM FLOW (FROM LOG SOURCES TO INVESTIGATION)

1 LOG SOURCES

- Servers (Windows/Linux)
- Endpoints
- Firewalls
- Cloud Services (AWS, Azure, GCP)
- Applications (Web, Database)
- Network Devices (Routers, Switches)
- Identity Systems (Active Directory, Entra ID)

2 COLLECTION

Logs are collected using agents, Syslog or APIs and sent securely to the SIEM.

3 PROCESSING

Logs are parsed, normalized and indexed so they can be searched and analyzed.

4 DETECTION

The SIEM looks for suspicious patterns using rules, correlation and threat intelligence.

5 ALERTS

When a potential threat is found the SIEM generates an alert.

6 INVESTIGATION

Security teams investigate the alert, confirm if it is a real threat and take action.

4. EXAMPLE SCENARIO

1. A user enters the wrong password multiple times.
2. The server creates authentication logs.
3. Logs are collected and sent to the SIEM.
4. The SIEM detects multiple failed logins.
5. An alert is generated for suspicious activity.
6. The security team investigates and checks the user, IP address and time.
7. If confirmed malicious, they take action such as blocking the IP or disabling the account.

5. KEY COMPONENTS INVOLVED

- Log Collectors (Agents/Syslog/API)
- Parsing and Normalization
- Indexing and Storage
- Detection Rules and Correlation
- Dashboards and Visualization
- Alerting and Notifications
- Threat Intelligence (Optional)
- User and Entity Monitoring (Optional)
- Incident Response and Investigation

6. IMPORTANT TO REMEMBER

- SIEM does not block attacks automatically.
- It provides visibility so securiity teams can act.
- Good log collection and quality data are essential.
- It works best when combined with people, processes and other security tools.

7. JUNIOR ENGINEER TIP

Start by understanding the log flow. If logs are not being collected or are incomplete, the SIEM will miss important security events.

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Understanding Security Logs

1. WHAT ARE SECURITY LOGS?

Security logs are **records of events** that happen on systems, applications, networks and security devices. They capture user activity, system activity, security events and any changes that occur.

2. WHAT DO LOGS RECORD?

- Successful and failed login attempts
- User actions (file access, permission changes)
- System events (startup, shutdown, errors)
- Application activity (web requests, database activity)
- Network activity (connections, blocked traffic)
- Security events (firewall alerts, malware detection)
- Configuration changes (settings, policy updates)

3. WHERE DO SECURITY LOGS COME FROM?



Servers
- Linux logs
- Windows event logs



Endpoints
- Laptops
- User devices



Firewalls
- Traffic logs
- Blocked requests



Cloud Services
- AWS, Azure, GCP
- Audit logs



Applications
- Web servers
- Database logs



Network Devices
- Routers
- Switches

4. EXAMPLE SECURITY LOG ENTRIES

TIMESTAMP	SOURCE	EVENT	DESCRIPTION
2024-01-10 10:15:22	Linux Server	SSH login	User devops logged in from 192.168.1.10
2024-01-10 10:16:03	Firewall	Blocked	Blocked connection from 203.0.113.5
2024-01-10 10:17:45	Windows	Login failed	Failed login for user admin
2024-01-10 10:18:21	Web Server	HTTP request	GET /login from 198.51.100.23
2024-01-10 10:19:10	Database	Query	SELECT * FROM users

5. WHY ARE LOGS IMPORTANT?

- Help detect suspicious and malicious activity
- Provide evidence during security investigations
- Show what happened, when it happened and who did it
- Help with incident response and root cause analysis
- Support compliance and audit requirements
- Give visibility into system and network activity
- Help identify misconfigurations and policy violations

6. KEY POINTS TO REMEMBER

- Logs tell the story of what is happening in your environment.
- Different systems generate different types of logs.
- Logs are essential for detecting, investigating and responding to security threats.
- Good log collection and retention make security teams more effective.

7. JUNIOR ENGINEER TIP

Whenever troubleshooting a security issue, always check the relevant logs. Logs often contain the exact evidence you need to understand what happened.

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Common Security Log Sources

1. WHAT ARE LOG SOURCES?

Log sources are systems and applications that generate log data about their activity. These logs are collected by the SIEM to give visibility into what is happening across your entire environment.

2. COMMON LOG SOURCE CATEGORIES

- Operating systems (Windows, Linux)
- Network and security devices (firewalls, routers)
- Cloud services (AWS, Azure, GCP)
- Applications (web, database, business apps)
- Identity systems (Active Directory, Entra ID)
- Infrastructure and virtualization (VMware, Hyper-V)

3. EXAMPLES OF COMMON SECURITY LOG SOURCES

Windows

- Windows Event Logs
- Login and authentication
- User and group changes
- System events
- Security policy changes

Linux

- System logs (syslog)
- Authentication logs (auth.log)
- User activity
- Sudo and privilege changes
- System and service events

Firewalls

- Traffic logs (allowed/blocked)
- Connection attempts
- Intrusion and threat events
- VPN activity
- Policy changes

Cloud Services

- AWS CloudTrail
- Azure Activity Logs
- GCP Audit Logs
- Resource changes
- Identity and access activity

Applications

- Web server logs (e.g. Nginx, Apache)
- Application logs
- Database logs (e.g. MySQL, PostgreSQL)
- API and user activity
- Error and access logs

Identity Systems

- Active Directory logs
- Entra ID (Azure AD) logs
- User login and logout
- Account and group changes
- Failed authentication attempts

4. WHY THESE LOG SOURCES MATTER

- Give a complete view of activity across your environment
- Help detect suspicious and malicious behavior
- Provide evidence for investigations
- Show who did what, when and from where
- Help meet security and compliance requirements

5. JUNIOR ENGINEER TIPS

- Start with key sources (Windows, Linux, firewalls)
- Ensure logs have accurate time (use NTP)
- Collect both successful and failed events
- Work with your security team to understand which logs are most important
- Check that logs are being received in the SIEM

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Events, Alerts, and Incidents

1. OVERVIEW

Events, alerts and incidents represent different leverls of security activity. Events are recorded activities, alerts are potentially suspicious activities, and incidents are confirmed security threats that require action.

2. KEY DIFFERENCES

EVENT

- Recorded activity that happens on a system
- Can be normal or suspicious
- Examples: login, file access, configuration change

ALERT

- Triggered when an event matches a detection rule
- Indicates potentially suspicious activity
- Needs review to confirm if it is a real threat

INCIDENT

- Confirmed security threat
- Validated after investigation
- Requires response actions to contain, remove or mitigate the threat

3. REAL WORLD EXAMPLES

Scenario	Event (Recorded Activity)	Alert (Suspicious Activity)	Incident (Confirmed Threat)
User Login	User login from 192.168.1.10	Multiple failed logins from the same IP	Confirmed brute force attack on user account
File Activity	File accessed on a server	Sensitive file accessed by an unusual user	Unauthorized access to sensitive data
Network Activity	Connection to a remote IP	Connection to a known malicious IP address	Confirmed malware command and control activity
System Changes	Firewall rule modified	Unexpected change to critical firewall rule	Malicious change that exposed the network
Application Activity	New admin user created	Admin user created outside business hours	Compromised account used to create unauthorized admin

4. HOW THEY ARE RELATED

Event → Activity is recorded in logs

Alert → Event matches a detection rule and triggers an alert

Incident → After investigation, it is confirmed as a real security threat and requires action

5. KEY POINTS TO REMEMBER

- All alerts are based on events
- Not all events are suspicious
- An alert is a signal, not a confirmed threat
- An incident is a validated security threat
- Incidents require investigation and response
- Good detection rules help reduce false alerts

6. COMMON ALERT TYPES

- Multiple failed login attempts
- Login from an unusual location
- Access to sensitive files
- Unusual process or service activity
- Malware detected
- Suspicious network connections
- Privilege escalation attempts
- Changes to security configurations

7. JUNIOR ENGINEER TIP

Always investigate an alert before assuming it is a real incident. Look at the full context, including user, time, source IP, affected system and related events.

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How Logs Reach a SIEM

1. OVERVIEW

Logs can reach a SIEM using agents, Syslog, APIs or collectors. These methods collect log data from different systems and send it securely to the SIEM for processing, analysis and detection of threats.

2. COMMON METHODS

Agents	Small software installed on systems that collect and forward logs.
Syslog	Standard protocol used to send logs over the network.
APIs	Used by cloud services and applications to send logs to the SIEM.
Collectors	Central components that receive logs from multiple sources and forward them to the SIEM.

3. HOW LOGS REACH A SIEM (FLOW DIAGRAM)

Log Sources

- Servers (Windows/Linux)
- Endpoints (Laptops/Devices)
- Firewalls
- Cloud Services (AWS, Azure, GCP)
- Applications (Web, Database)
- Network Devices (Routers, Switches)
- Identity Systems (Active Directory, Entra ID)

→

Collection Methods

- **Agents** (Installed on hosts)
- **Syslog** (Sends logs over network)
- **APIs** (Pulls or pushes logs)
- **Collectors** (Receives logs from multiple sources)

→

Secure Log Forwarding

- Logs are encrypted during transit (e.g. TLS).
- Authentication is used (API keys, certificates).
- Firewall rules and access controls restrict who can send logs.

→

SIEM Platform

- Receives logs
- Parses and normalizes data
- Stores logs
- Analyzes events
- Runs detection rules
- Generates alerts
- Provides dashboards and search

4. EXAMPLE LOG COLLECTION METHODS

Method	Common Use Cases	Example Sources
Agent	Servers, endpoints	Windows, Linux
Syslog	Network devices, firewalls	Firewalls, routers, switches
API	Cloud services, SaaS apps	AWS CloudTrail, Microsoft 365, Google Workspace
Collector	Multiple sources	Syslog collectors, log forwarders (e.g. Fluentd, Logstash, Beats)

5. KEY POINTS TO REMEMBER

- Different systems use different methods to send logs.
- Agents collect logs directly from hosts.
- Syslog is widely used for network devices and Linux systems.
- APIs are common for cloud services and applications.
- Collectors help gather logs from many sources.
- Logs should be sent securely using encryption and authentication.

6. COMMON CHALLENGES

- Logs not being sent due to network or firewall issues
- Incorrect agent configuration
- Missing permissions for API access
- Inconsistent log formats from different sources

7. JUNIOR ENGINEER TIP

Always verify that logs are being received in the SIEM. Check timestamps, source information and the expected events to confirm that the log forwarding method is working correctly.

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Parsing and Normalizing Logs

1. WHAT IS LOG PARSING?

Parsing is the process of reading a log entry, identifying the different parts (fields) and extracting important information such as time, source, user, IP address, event type and action.

2. WHAT IS LOG NORMALIZATION?

Normalization means converting different log formats into a standard structure so they can be easily searched, correlated and analyzed in the SIEM.

- Different systems write logs in different formats.
- Normalization converts them into a common format.
- All important fields are mapped to the same names.
- This makes search, detection and correlation easier.

3. EXAMPLE: FROM RAW LOG TO NORMALIZED EVENT

RAW LOG (DIFFERENT FORMATS)

Windows (Event Log)	EventID=4624 AccountName=jsmith IpAddress=192.168.1.10 LogonType=3
Linux (Syslog)	Jan 10 10:15:22 server1 sshd[1234]: Accepted password for jsmith from 192.168.1.10 port 54321
Firewall (Vendor Format)	<190>Jan 10 10:15:22 FW01 %ASA-6-302013: Built outbound TCP connection 192.168.1.10/54321 to 8.8.8.8/80

PARSING (EXTRACT FIELDS)

- Identify log format
- Use parsing rules or patterns
- Extract fields (e.g. time, user, IP)
- Handle different delimiters and structures
- Convert to key-value fields

NORMALIZED EVENT (STANDARD FORMAT)

```
{
  "timestamp": "2024-01-10T10:15:22",
  "source": "server1",
  "event_type": "login_success",
  "user": "jsmith",
  "source_ip": "192.168.1.10",
  "destination_ip": "-",
  "device_type": "linux_server",
  "log_source": "auth",
  "action": "accepted_password"
}
```

4. COMMON LOG FORMATS

Format	Description	Example
Syslog	Common format for Linux, network devices	Jan 10 10:15:22 server1 sshd[1234]: message
Windows Event Log	Windows system and security events	EventID=4624 AccountName=jsmith ...
JSON	Structured format used by many tools	{"time":"2024-01-10", "event": "login"}
CEF	Used by many security devices	CEF:0|Vendor|Product|...
CSV	Simple comma separated format	2024-01-10,jsmith,login,...
Custom/Proprietary	Vendor specific format	Varies by tool

5. KEY BENEFITS

- Turns different log formats into a standard structure
- Makes logs searchable and filterable
- Enables correlation across different sources
- Improves detection accuracy
- Reduces false positives
- Helps with dashboards, alerts and reporting
- Works with logs from many systems and vendors

6. COMMON CHALLENGES

- Different log formats and structures
- Missing or inconsistent fields
- Custom vendor formats
- Parsing errors due to format changes
- Large volume of logs

7. JUNIOR ENGINEER TIP

When setting up log parsing, always test with real log samples from each source, Verify that important fields such as timestamp, user, IP address and event type are being extracted correctly.

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Understanding Log Fields

1. WHAT ARE LOG FIELDS?

Log fields are individual pieces of information in a log entry that describe an event. Each field provides context such as time, user, source, destination, event type, and the action that occurred.

2. COMMON LOG FIELDS

- **Timestamp** — When the event happened
- **Username** — Which user performed the action
- **Source IP Address** — Where the event came from
- **Destination IP Address** — Where the event is going
- **Event ID** — Unique ID for the event type
- **Action** — What happened during the event

3. EXAMPLE LOG ENTRY AND FIELDS

`2024-01-10 10:15:22` `server1` `192.168.1.10` `203.0.113.5` `4624` `User login successful`

- **Timestamp** — When the event occurred
- **Hostname** — Which system generated the log
- **Source IP Address** — Where the event came from
- **Destination IP Address** — Where the event is going to
- **Event ID** — Identifier for the event type
- **Action** — What happened during the event

4. COMMON LOG FIELDS AND MEANINGS

Field	Description	Example	What It Tells You
Timestamp	Date and time of the event	2024-01-10 10:15:22	When the event happened
Username	User associated with the event	jsmith	Who performed the action
Source IP Address	Origin of the event	192.168.1.10	Where the event came from
Destination IP Address	Target of the event	203.0.113.5	Where the event is going
Event ID	Unique identifier for event type	4624	What type of event it is
Action	Description of what happened	User login successful	The action that occurred

5. KEY POINTS TO REMEMBER

- Every log entry contains multiple fields.
- Fields give important context about the event.
- Timestamps help you build a timeline.
- Source and destination IPs show where activity is coming from and going to.
- Event IDs identify the type of event.
- Actions tell you what actually happened.

6. COMMON MISTAKES

- Ignoring important fields like timestamp or source IP
- Misinterpreting the meaning of event IDs
- Not checking all relevant fields together
- Assuming a single field tells the full story
- Overlooking unusual values in log fields

7. JUNIOR ENGINEER TIP

Always look at multiple fields together. A single field can be normal, but when combined with other fields, it can reveal suspicious activity.

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Searching Security Events

1. WHAT IS LOG SEARCHING?

Log searching allows security analysts to filter and find specific events in large volumes of log data. It helps identify suspicious activity, troubleshoot issues and investigate security incidents.

2. COMMON SEARCH FILTERS

Time Range
Search within a specific date and time period

Username
Find events for a specific user

Source IP
Filter by the originating IP address

Destination IP
Find events going to a specific IP address

Event ID
Search for a specific event identifier

Event Type/Action
Filter by action type (login, file access, process start, etc.)

3. EXAMPLES OF SEARCHING SECURITY EVENTS

Use Case	Example Query (Generic Syntax)	What It Finds
Find failed logins	event_type=login AND status=failed	All failed login attempts
Find activity from a specific user	user=jsmith	All events for user jsmith
Find events from a source IP	source_ip=192.168.1.10	All events from this IP address
Find events to a destination IP	destination_ip=10.0.0.5	All events going to this IP address
Find a specific event ID	event_id=4624	All events with ID 4624 (e.g. successful logon)
Find dangerous actions	action IN ("file_delete", "process_start")	Events related to file deletion or process start
Find events in a time range	timestamp>=2024-01-01 AND timestamp<=2024-01-02	All events within the selected time period

4. SEARCH EXAMPLE: FIND SUSPICIOUS ACTIVITY

Example Query

```
event_type=login AND status=failed
AND source_ip=203.0.113.5
```

Sample Search Results

Timestamp	User	Source IP	Event ID	Action	Status
2024-01-10 10:15:22	admin	203.0.113.5	4625	Login	Failed
2024-01-10 10:16:05	admin	203.0.113.5	4625	Login	Failed
2024-01-10 10:17:31	admin	203.0.113.5	4625	Login	Failed
2024-01-10 10:18:02	admin	203.0.113.5	4625	Login	Failed

5. KEY POINTS TO REMEMBER

- Use filters to narrow down large volumes of logs
- Combine multiple fields for accurate results
- Look for patterns and unusual activity
- Adjust time range based on the investigation
- Verify results to confirm if it is a real threat

6. COMMON MISTAKES

- Using too broad search terms
- Not setting the correct time range
- Misspelling field names
- Ignoring relevant fields
- Assuming one event is a real threat
- Not reviewing the full context

7. JUNIOR ENGINEER TIP

Start with simple queries and use one filter at a time. Once you understand the results, add more filters to narrow down the search and find the exact events you need.

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Building Your First SIEM Dashboard

1. WHAT IS A SIEM DASHBOARD?

A SIEM dashboard is a visual representation of security data that shows key information such as event counts, trends, alerts and system activity. It helps security teams quickly understand what is happening in their environment.

2. COMMON DASHBOARD ELEMENTS

- **Event Counts** – Total number of events over time
- **Trends** – Increase or decrease in activity
- **Alerts** – Recent and high priority alerts
- **Visualizations** – Charts, graphs and breakdowns of log data
- **System Monitoring** – Health and status of key systems
- **Top Activity** – Top users, IP addresses, event types and sources

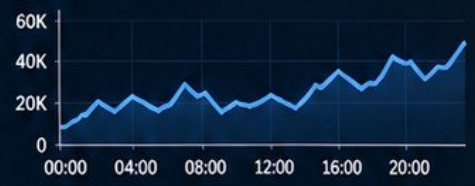
3. EXAMPLE SIEM DASHBOARD (SAMPLE VIEW)

Security Overview

Last 24 Hours

Total Events	Total Alerts	Critical Alerts	Unique Users
1,245,672	428	56	1,183
↑ 12%	↑ 35%	↑ 20%	↑ 8%

Events Over Time



Events by Type



Authentication	35%
File Access	20%
Network	18%
System	15%
Other	12%

Top Source IP Addresses

192.168.1.10	4,320
203.0.113.5	3,890
10.0.0.5	2,540
172.16.0.8	1,920
198.51.100.2	1,430

Top Users by Activity

jsmith	12,430
admin	8,320
mjones	5,620
svc_backup	3,410
dbadmin	2,980

Top Event Categories

Successful Login	28,432
Failed Login	12,903
File Access	10,421
Process Start	8,542
Configuration Change	5,321

Geographic Source (Example)



North America	45%
Europe	25%
Asia	20%
Other	10%

4. COMMON VISUALIZATION TYPES

- Line charts – show event trends over time
- Bar charts – compare event types, users or sources
- Pie charts – show distribution of events
- Tables – display detailed event information
- Maps – show source locations (geographic data)

5. KEY POINTS TO REMEMBER

- Dashboards give a quick overview of security activity
- Visualizations help identify unusual patterns
- Monitor key metrics such as total events and alerts
- Include top users, source IPs and event types
- Keep the dashboard simple and relevant
- Use real time or near real time data when possible

6. COMMON MISTAKES

- Adding too many visualizations
- Monitoring unimportant data
- Using the wrong time range
- Ignoring critical alerts
- Making the dashboard too complex
- Not updating the dashboard regularly

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How SIEM Detects Threats

1. WHAT IS DETECTION?

Detection in a SIEM means finding **suspicious activity** from logs using **rules, correlation and analytics.** It helps identify real threats such as attacks, insider risks and malicious behavior.



2. KEY DETECTION METHODS

Detection Rules
Predefined rules to spot known threats.

Correlation
Link related events across sources.

Behavior Analytics
Find unusual patterns and risky behavior.

Indicators of Compromise
Detect known signs of attack (IOCs).

3. EXAMPLE: DETECTION FLOW

1. **Logs Collected** — Events from endpoints, apps and network.
2. **Rules Applied** — SIEM checks against detection rules.
3. **Correlation Performed** — Related events are linked together.
4. **Suspicious Activity Found** — Unusual behavior is detected.
5. **Alert Generated** — Analyst is notified with details.

4. COMMON THREATS DETECTED

- Brute Force Login Attempts
- Privilege Escalation
- Lateral Movement
- Data Exfiltration
- Malware Activity
- Suspicious Network Connections
- Policy Violations



5. WHAT IS A CORRELATION RULE?

A correlation rule combines multiple events to detect bigger threats. For example, many failed logins from different IP addresses can indicate a brute force attack, or a login from a new country followed by data access can indicate a compromised account.



6. KEY TAKEAWAYS

- SIEM detects threats using rules, correlation and behavior analysis.
- It finds suspicious activity, not just single events.
- Indicators of compromise help identify known attacks.
- Early detection enables faster response and reduces damage.



Detect Analyze Respond

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Detecting Failed Login Attempts

1. WHAT IT MEANS

Repeated authentication failures can indicate a **password attack**, a **brute force attempt**, or a **configuration problem** such as a wrong password policy or blocked account.



2. COMMON EXAMPLES

Brute Force Attack
Many login attempts with different passwords.

Wrong Passwords
User keeps entering the wrong password.

Configuration Issue
Account lockout or policy blocking valid users.

3. DETECTION INDICATORS

Common Signs

- Multiple failed logins from the same IP
- Repeated failures for one username
- Login attempts at unusual times
- Failed logins from different locations
- Spike in authentication failures

Example Log Entries

```
2024-10-10 08:15:21 user=john status=failed ip=203.0.113.45
2024-10-10 08:15:32 user=john status=failed ip=203.0.113.45
2024-10-10 08:15:45 user=john status=failed ip=203.0.113.45
2024-10-10 08:16:10 user=john status=failed ip=203.0.113.45
```

Many failures in a short time is a **red flag.**



4. ANALYSIS STEPS

1. Check the number of failed attempts
2. Look for the affected user account
3. Identify the source IP address
4. Check for unusual locations or times
5. Determine if it is an attack or mistake

5. MITIGATION ACTIONS

- Block or rate limit suspicious IP addresses
- Enforce strong password policies
- Enable multi factor authentication
- Monitor for repeated failures
- Alert the security team for investigation

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Understanding SIEM Correlation Rules

1. WHAT ARE CORRELATION RULES?

Correlation rules combine **multiple related events** from different sources to identify potentially malicious activity. Instead of looking at a single log entry, they look for **patterns and connections** between events to detect threats that may not be obvious on their own.

2. WHY CORRELATION MATTERS

Detects complex attacks
Finds threats that span multiple systems and events.

Reduces false positives
Looks for meaningful patterns instead of single random events.

Identifies stealthy activity
Helps detect attacks that try to avoid simple detection rules.

Provides better context
Shows the full picture of what is happening across your environment.

3. EXAMPLE OF A CORRELATION RULE

Event 1
Multiple failed login attempts from the same IP
(10:00 AM)

→

Event 2
Successful login from the same IP
(10:05 AM)

→

Event 3
Access to sensitive files or admin tools
(10:10 AM)

→

Event 4
Large data transfer to an external IP
(10:15 AM)

→

Correlation Result
Potential account compromise or data exfiltration detected

4. COMMON CORRELATION RULES

- Multiple failed logins followed by a successful login
- Login from unusual location followed by sensitive access
- User account creation followed by privilege escalation
- Suspicious process execution followed by network activity
- Data access followed by large data transfer
- Changes to security settings followed by service restart

5. KEY POINTS TO REMEMBER

- Correlation rules connect multiple events.
- They help detect complex and stealthy threats.
- Rules use time, user, IP, device and event type relationships.
- They reduce false positives by looking at patterns.
- Correlation provides better context for investigations.
- You can customize rules based on your environment and risks.

6. REAL WORLD EXAMPLE (ATTACK CHAIN)

Brute force login attempts → Successful login → Privilege escalation → Access to sensitive data → Data exfiltration

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False Positives and Alert Fatigue

1. WHAT ARE FALSE POSITIVES?

A false positive is an alert that **looks like a security threat** but is actually **legitimate and harmless activity**. They can be caused by normal user behavior, expected system processes or misconfigured rules.

EXAMPLE

An alert for multiple failed logins, but the user simply typed the wrong password several times.

2. WHAT IS ALERT FATIGUE?

Alert fatigue happens when analysts receive **too many alerts**, including many false positives. It becomes difficult to focus on real threats, which can lead to slower response times and **missed attacks**.



3. COMMON CAUSES OF FALSE POSITIVES

Legitimate user activity
Valid actions that look suspicious (e.g. many logins)

System processes
Backups, updates or scheduled tasks triggering alerts

Misconfigured rules
Rules that are too broad or too sensitive

Unexpected environment changes
New devices, new locations or network changes

4. IMPACT OF ALERT FATIGUE

- Analysts may miss real threats
- Slower incident response
- Higher stress and burnout
- Lower confidence in alerts
- Important security issues may go unnoticed

5. HOW ANALYSTS IMPROVE DETECTION QUALITY

Tune detection rules
Adjust thresholds and add better conditions

Use filters and exclusions
Exclude known legitimate activity (e.g. internal IPs)

Add context
Use user, device and environment information

Monitor and review
Regularly review alerts and refine rules based on real data

6. KEY TAKEAWAYS

- False positives are normal in SIEM.
- They happen due to legitimate activity, misconfigurations or environment changes.
- Alert fatigue can lead to missed threats.
- Analysts improve detection quality by tuning rules, using filters and adding context.
- A well tuned SIEM provides higher quality alerts and better security outcomes.

7. EXAMPLE: FALSE POSITIVE VS REAL THREAT

FALSE POSITIVE

- Multiple failed logins from a known user during a password reset
- Large file download by a backup system
- Login from a new device that is company issued

REAL THREAT

- Multiple failed logins followed by a successful login from an unknown IP
- Access to sensitive files from an unusual location
- Unexpected admin account activity outside normal hours

VERIQTA

Investigating a Security Alert

1. WHAT IS A SECURITY ALERT?

A security alert is a notification generated by a SIEM when suspicious or potentially malicious activity is detected in your environment.

It requires investigation to determine if it is a real threat or a false positive.

2. INVESTIGATION WORKFLOW

Review the alert → **Check key details** → **Analyze related logs** → **Determine the outcome**

- **Review the alert**: Understand what triggered it and the severity.
- **Check key details**: Look at timestamps, users, IP addresses and affected systems.
- **Analyze related logs**: Find more evidence across multiple sources.
- **Determine the outcome**: Confirm if it is a real threat or a false positive and take action.

3. KEY INFORMATION TO INVESTIGATE

- **Timestamps**: When did the event happen? Check the full timeline.
- **Users**: Which user performed the action? Was it expected behavior?
- **IP Addresses**: Where did the traffic come from? Is the IP address known or suspicious?
- **Logs**: Check logs from different sources (e.g. authentication, system, network, application).
- **Supporting Evidence**: Look for related events, file changes, process activity or network connections to understand the full picture.

4. EXAMPLE INVESTIGATION

10:02 AM	**SIEM alert triggered** Multiple failed login attempts detected for user john.doe.
10:02–10:05 AM	**Check user activity** Confirm this user does not normally login from this IP.
10:03 AM	**Investigate IP address** IP address 185.220.101.5 identified as a high risk location.
10:04 AM	**Review additional logs** Found successful login after multiple failures.
10:05 AM	**Check related activity** Suspicious file download and new process execution.
10:10 AM	**Conclusion** Confirmed as a real threat. Account disabled and incident escalated.

5. INVESTIGATION QUESTIONS

- Is the activity expected for this user?
- Is the IP address known or suspicious?
- Are there multiple related events?
- What other systems show the same activity?
- Is there any data access or file download?
- Does the behavior match known attack patterns?
- Is this a false positive or a real threat?

6. OUTCOMES AND ACTIONS

- **Real Threat**: Take immediate action (e.g. block IP, disable account, escalate to incident response).
- **False Positive**: Document the reason and tune detection rules to reduce similar alerts.
- **Needs More Review**: Gather more evidence and continue investigation.
- **Improve Detection**: Use lessons learned to refine rules and reduce alert fatigue.

VERIQTA

Setting Up a SIEM Lab

Build a beginner-friendly, isolated SIEM environment using Wazuh and a Linux virtual machine.

1. WHAT IS WAZUH?

Wazuh is an open source SIEM and XDR platform that collects, analyzes and visualizes security events from your systems.

wazuh

- **Collects logs and events**
- **Detects threats and suspicious activity**
- **Provides dashboards and alerts**
- **Open source and free to use**

2. LAB ARCHITECTURE

Linux VM (Agent)
- Generate logs
- Send events to Wazuh

Agent sends logs →

Wazuh Server (Manager + Indexer + Dashboard)
- Collects logs
- Analyzes events
- Stores data
- Provides dashboards

→

Wazuh Dashboard
- View alerts
- Search logs
- Monitor security events

3. REQUIREMENTS

Virtualization Software
- VirtualBox
- VMware
- or a cloud VM

Wazuh Server VM
- Ubuntu Server
- Minimum 4GB RAM
- 2 vCPUs
- 50GB Storage

Linux Agent VM
- Ubuntu or other Linux distro
- Minimum 2GB RAM
- 2 vCPUs
- 20GB Storage

Network Setup
- Isolated lab environment
- Internet access for initial setup

4. HIGH LEVEL SETUP STEPS

1. Create virtual machines for Wazuh server and Linux agent.
2. Install Wazuh server (manager, indexer and dashboard).
3. Configure the Linux agent to send logs to the server.
4. Access the Wazuh dashboard in your browser.
5. Generate logs on the agent and verify events in Wazuh.

5. WHAT YOU WILL HAVE



6. KEY BENEFITS

- Hands on experience with a real SIEM.
- Learn how to collect and analyze logs.
- Understand security alerts and threats.
- Practice in a safe and isolated environment.
- Build a strong foundation for real world security monitoring.

VERIQTA

Connecting Linux Logs to SIEM

Install a monitoring agent, collect Linux authentication events, and verify that logs reach the SIEM.

1. OVERVIEW

We will install the Wazuh agent on a Linux machine to collect authentication logs (e.g. SSH logins) and send them to the Wazuh server for analysis.



2. AGENT INSTALLATION

1. Download and install the Wazuh agent on the Linux machine.
2. Register the agent with the Wazuh server using the server IP.
3. Start the agent service.
4. Verify the agent is connected to the server.

3. COLLECTING LINUX AUTHENTICATION LOGS

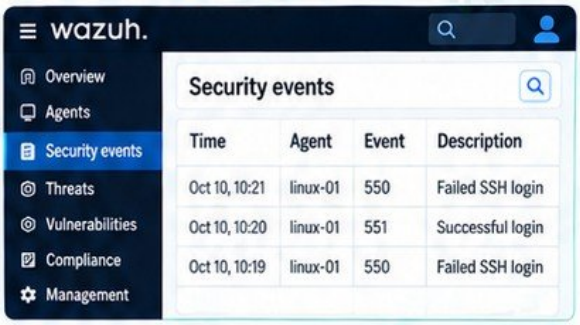
The Wazuh agent collects logs from /var/log/auth.log (Ubuntu) or /var/log/secure (RHEL/CentOS) which contain authentication events such as:

/var/log/auth.log

- Successful logins
- Failed login attempts
- SSH access
- User session events
- Privilege escalation attempts

4. VERIFY LOGS IN SIEM

Open the Wazuh dashboard and check that the Linux agent is sending logs. You should see authentication events appear in real time.



5. EXAMPLE AUTHENTICATION EVENTS

```
Oct 10 10:15:12 linux-01 sshd[1234]: Failed password for
invalid user admin from 192.168.1.50 port 54321 ssh2
Oct 10 10:16:05 linux-01 sshd[1250]: Accepted password for
ubuntu from 192.168.1.25 port 43211 ssh2
Oct 10 10:17:33 linux-01 sudo:   ubuntu : TTY=pts/0 ;
USER=root ; COMMAND=/bin/systemctl
Oct 10 10:18:21 linux-01 sshd[1301]: Failed password for
root from 203.0.113.10 port 61322 ssh2
```

6. TEST AND GENERATE LOGS

Generate Failed Login
Attempt to SSH with a wrong password.

Generate Successful Login
Login with a valid user account.

Check Logs on Linux
View logs using tail -f /var/log/auth.log

Verify in SIEM
Confirm events appear in the Wazuh dashboard in real time.

7. KEY TAKEAWAYS

- The Wazuh agent collects Linux authentication logs automatically.
- You can monitor successful and failed logins in real time.
- Logs help detect suspicious activity such as brute force attacks.
- Authentication logs provide important evidence for investigations.
- This setup gives you hands on experience with a real SIEM.